Elementary Matrices

* **Lemma 1.1** Matrix multiplication is associative ie $A.(B.C) = (A.B).C$

* **Lemma 1.2** If $P = \left[\begin{array}{c|c} A & B \\ \hline C & D \end{array}\right]$, $Q = \left[\begin{array}{c|c} A' & B' \\ \hline C' & D' \end{array}\right]$ are matrices s.t (i) $\exists R = P.Q$ and (ii) # columns of A = # rows of A'

  then $R = \begin{bmatrix} AA' + BC' & AB' + BD' \\ CA' + DC' & CB' + DD' \end{bmatrix}$

* **Defn 1.3** A matrix A is called invertible if $\exists\ B$ s.t $A.B = B.A = I$

* **Lemma 1.4** A is invertible $\iff$ A^T is invertible

* **Lemma 1.5** If A, B and C are matrices s.t $C = A.B$ and both A and B are invertible $\Longrightarrow$ C is invertible

* **Lemma 1.6** Elementary Matrices $\begin{bmatrix} 1 & c \\ 0 & 1 \end{bmatrix} / \begin{bmatrix} 1 & 0 \\ c & 1 \end{bmatrix}$, $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$, $\begin{bmatrix} c & 0 \\ 0 & 1 \end{bmatrix} / \begin{bmatrix} 1 & 0 \\ 0 & c \end{bmatrix}$

  → Effect of left and right multiplication

  → They are invertible

* **Defn 1.7** Reduced row echelon form (rref):

  ① The first non zero element in a row, ie the pivot, is 1. All the elements above and below a pivot are zero

  ② If row i and row j are two rows of rref s.t (i) $j > i$ and (ii) they both have pivot elements then row j pivot lies to the right of row i pivot

  ③ If row i is all zeros then all the rows below it are also all zeros.

* **Lemma 1.8**

  A (square matrix)

  ↙ invertible — non invertible ↘

  $\text{rref}(A) = I$; $\text{rref}(A) = \begin{bmatrix} P & Q \\ 0 & 0 \end{bmatrix}$ and also $\exists$ an invertible matrix M s.t $\text{rref}(A) = \begin{bmatrix} I & 0 \\ 0 & 0 \end{bmatrix} M$

* **Theorem 1.9** If A, B, C are square matrices s.t $C = A.B$

  C is invertible $\iff$ Both A and B are invertible

  Hint Use 1.8 and 1.5

Determinant Map.

* Lemma 1.10 — A is an invertible matrix $\iff$ A is product of elementary matrices

* Defn 1.11 — Determinant : $\mathbb{R}^{n\times n} \longrightarrow \mathbb{R}$ s.t

① $\det(I) = 1$

② $\det\begin{pmatrix}\vdots\\ aP+bQ\\ \vdots\end{pmatrix} = a\det\begin{pmatrix}\vdots\\ P\\ \vdots\end{pmatrix} + b\det\begin{pmatrix}\vdots\\ Q\\ \vdots\end{pmatrix}$, ie det is linear in rows

③ If two rows of A are same then $\det(A) = 0$

* Lemma 1.12 — If E is an elementary matrix and A are square matrix then

$$\det(EA) = \det(E)\det(A)$$

Hint: 1.6, 1.11

* Theorem 1.13 — If A and B are square matrices of same size then

$$\det(AB) = \det(A)\det(B)$$

Hint 1.12, 1.10

* Lemma 1.14 — If A is a square matrix then $\det(A) = \det(A^T)$

* Theorem 1.15 — If A, P, Q are square matrices s.t $A = \begin{bmatrix} P & 0 \\ 0 & Q \end{bmatrix}$ then

$$\det(A) = \det(P)\det(Q)$$

Hint (i) P or Q is non invertible

(ii) P and Q are invertible

$f(B) := \det\begin{bmatrix} B & 0 \\ 0 & Q \end{bmatrix} \Big/ \det(Q)$

$f(B)$ satisfies 1.11 and $\therefore$ a det map.

* Note 1.16 — Discussions from 1.1 to 1.15 can be extended to complex matrices.

* Defn 1.17 — S_n: Symmetric group: Group of permutations of n objects

Cyclic notation for elements of S_n

* Defn 1.18 — GL_n: General linear group

$$GL_n := \{ A \in \mathbb{R}^{n\times n} \text{ s.t } AB = I \text{ where } B \in \mathbb{R}^{n\times n} \}$$

* Defn 1.19 SL_n : Special linear group

$$SL_n := \{ A \in \mathbb{R}^{n \times n} \text{ s.t } \det(A) = 1 \}$$

* Defn 1.20 O_n : Orthonormal group

$$O_n := \{ A \in \mathbb{R}^{n \times n} \text{ s.t } A \cdot A^T = A^T \cdot A = I \}$$

* Defn 1.21 SO_n : Special Orthonormal group

$$SO_n := \{ A \in \mathbb{R}^{n \times n} \text{ s.t } A \cdot A^T = I \text{ and } \det(A) = 1 \}$$

* Lemma 1.22 Let A and B are matrices and x a column vector s.t.

(i) $Ax = 0$

(ii) $BAx = 0$

(a) $\text{rank}(B) = \#\text{ columns of } B \implies \dim(\text{solution space of (i)}) = \dim(\text{solution space of (ii)})$

(b) $\text{rank}(B) < \#\text{ columns of } B \implies \dim(\text{solution space of (i)}) \leq \dim(\text{solution space of (ii)})$

Normal Groups

* **Defn 2.1** Law of composition on a set S is a function $f : \underbrace{S \times S}_{\text{cartesian product}} \to S$

* **Def 2.2** $e \in S$ is called identity w.r.t a law of composition on set S if $\forall a \in S \quad e.a = a.e$.
Caution: A set might not contain an identity element w.r.t a law of composition

* **Defn 2.3** $\tau \in S_n$ is called a transposition if it permutes only two objects.

* **Defn 2.4** Group: A set G together with a law of composition $(.)$ s.t
① $(a.b).c = a.(b.c) \quad \forall a, b, c \in G$
② $\exists e \in G$ s.t $ea = a.e \quad \forall a \in G$
③ $a \in G \Rightarrow \exists a^{-1} \in G$ s.t $a.a^{-1} = a^{-1}.a = e$

* **Example 2.5** S_3 : Symmetry group of 3 elements
$x = (123) \quad y = (12)$ generators
$x^3 = 1 \quad y^2 = 1 \quad yx = x^2y$ defining relations
$S_3 = \{1, x, x^2, y, xy, x^2y\}$

* **Defn 2.6** Subgroup: A non empty subset $H \subset G$ s.t
① if $a, b \in H$ and $c = a.b \Rightarrow c \in H$
② if $a \in H \Rightarrow a^{-1} \in H$

* **Example 2.7**
$$SO_n \subset SL_n \subset GL_n$$
$$C_n \subset GL_n$$

* **Defn 2.8** $H \subset G$ is called a normal subgroup of G, ie $H \trianglelefteq G$
if $\forall g \in G$ and $h \in H \quad ghg^{-1} \in H$

* **Defn 2.9** Order of a group G is defined as the number of unique elements in the group

Conjugates

* Defn 2.10 Let U be a subset of group G. Then the group generated by U is defined as the smallest subgroup H of G that contains all elements of U.

* Defn 2.11 Cyclic subgroup: The group generated by a single element x of a group

$$\langle x \rangle := \{ \ldots, x^{-2}, x^{-1}, e, x, x^2 \ldots \}$$

Order of x := Order of $\langle x \rangle$

* Defn 2.12 Group Homomorphism: $\varphi : G \longrightarrow G'$ s.t

$$\varphi(a.b) = \varphi(a).\varphi(b) \quad \forall\, a, b \in G$$

Image of $\varphi := \{ \varphi(g) : g \in G \}$

Kernel of $\varphi := \{ g \in G : \varphi(g) = e' \}$

* Lemma 2.13
(i) $\varphi(e) = e'$
(ii) $\varphi(a^{-1}) = \varphi(a)^{-1}$
(iii) if $H < G \implies \varphi(H) < G'$

* Lemma 2.14
(i) Kernel is a normal subgroup of G
(ii) Image is a subgroup of G'

* Lemma 2.15 Let $\varphi : G \longrightarrow G'$ be a group homomorphism, then

φ is injective $\iff$ Kernel of φ is $\{e\}$

* Defn 2.16 Isomorphism: A bijective group homomorphism

* Defn 2.17 Automorphism: An isomorphism from group to itself

* Defn 2.18 $a, b \in G$ are called conjugates if $\exists\, g \in G$ s.t $b = gag^{-1}$

Quotient Groups

* Defn 2.19 Let $H \subset G$ and $a \in G$ then

Left coset of $a := aH = \{a.h \text{ where } h \in H\}$

↳ All the left cosets wrt a subgroup have the same size

* Theorem 2.20 Let $H \subset G$ then $|G| = |H|\,(G:H)$

order of G | Order of H | # of left cosets

Hint: $a \underset{\text{related}}{R} b$ if $\exists h \in H$ s.t $a = b.h$ is an equivalence relation ⇒ Left cosets are non intersecting

$|G| = |H|\,|G:H|$ ⇐ 2.19

* Corollary 2.21 Let $\varphi : G \to G'$ be a group homomorphism,

$|G| = |\text{Kernel}|\,|\text{Image}|$

* Correspondence Theorem 2.22: Let $\varphi : G \to G'$ be a surjective group homomorphism

(i) $[H \subset G \text{ s.t } K \subset H] \xrightarrow[\text{bijective map}]{f} [\text{Subgroups of } G']$

(ii) If $\{H \lhd G \text{ and } K \subset H\} \xrightarrow[\text{bijective map}]{g} \{H' \lhd G\}$

* Hint: (a) We need $K \subset H$ (i) and (ii) for f and g to be injective

(b) we need φ to be surjective for (I) f, g to be onto

(II) $\varphi(H) \lhd G'$ where $H \lhd G$ and $K \subset H$

* Defn 2.23 Quotient Group: Let $H \lhd G$, $\bar{a} := aH$ left coset of $a \in G$ then

$G/H := \{\bar{a} : a \in G\}$ together with the law of composition $\bar{a}.\bar{b} = \overline{a.b}$ form a group

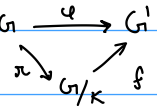
Further $\varphi : G \to G/H$ where $\varphi(a) = \bar{a}$ is a group homomorphism

Hint: Prove that the law of composition is independent of the choice of representative element and therefore valid. Then this law carries over the group structure.

Prime Number

* First Isomorphism Theorem 2.24

Let $\varphi: G \to G'$ be a surjective homomorphism and K be its kernel, then $f: G/K \to G'$ defined by $f(\bar{a}) := \varphi(a)$ is a group isomorphism.



* Lemma 2.25

Let W_1 and W_2 be subgroups of G, then the following are also subgroups of G

$$W_1 \cap W_2 := \{g \in G : g \in W_1 \text{ and } g \in W_2\}$$

$$W_1 + W_2 := \{g \in G : g = a.b \text{ where } a, b \in W_1 \cup W_2\}$$

Note: Usually $W_1 + W_2 \neq W_1 \cup W_2$

* Lemma 2.26

All the subgroups of $(\mathbb{Z}, +)$ have the form $n\mathbb{Z}$ where $n \in \mathbb{Z}$.

Hint: Use well ordering property of $\mathbb{Z}$ + Division by remainder

* GCD Defn 2.27

Let $a, b \in \mathbb{Z}$, then $d \in \mathbb{Z}$ is called gcd of a and b if

(i) $d > 0$

(ii) d divides a and b

(iii) if $c \in \mathbb{Z}$ divides a and b then c divides d

* Lemma 2.28 Let $a, b, d \in \mathbb{Z}$ then $d = g.c.d(a, b) \implies d\mathbb{Z} = a\mathbb{Z} + b\mathbb{Z}$

* Defn 2.29 $a, b \in \mathbb{Z}$ are called relatively prime if $g.c.d(a, b) = 1$

* Defn 2.30 Let $p \in \mathbb{Z}$ s.t if p divides $a.b \implies p$ divides either a or b

* Lemma 2.31 $p \in \mathbb{Z}$ is prime $\iff \nexists\ n \in \mathbb{Z}$ s.t $1 < n < p$ and n divides p.

L.C.M.

* L.C.M defn 2.32 Let $a, b \in \mathbb{Z}$, then $l \in \mathbb{Z}$ is called l.c.m of a and b if

(i) $l > 0$

(ii) a and b divides l

(iii) if a and b divides $c \in \mathbb{Z}$ then l divides c

* Lemma 2.33 Let $a, b, l \in \mathbb{Z}$ then $l = l.c.m(a,b) \iff l\mathbb{Z} = a\mathbb{Z} \cap b\mathbb{Z}$

* Lemma 2.34 $g.c.d(a,b) \times l.c.m(a,b) = a.b$

Proof (i) $a.b/d$ is divisible by l using 2.32

(ii) $l.d$ is divisible by $a.b$ using $d = ma + nb$

* Defn 2.35 Modular arithmetic $f: (\mathbb{Z}, *) \to (\mathbb{Z}/n\mathbb{Z}, *)$ where $n \in \mathbb{Z}$ is a group homomorphism (using 2.23)

further (i) $f(a+b) = f(a) + f(b)$

(ii) $f(a.b) = f(a).f(b)$

(iii) $f(a)(f(b) + f(c)) = f(a)f(b) + f(a)f(c)$

Congruence Relation $a \equiv b$ if $a-b$ is divisible by n

Vector Space

* Defn 3.1 A field F is a set together with two laws of composition, ie addition $(+)$ and multiplication $(\cdot)$ s.t

(i) $(F,+)$ is an abelian group with identity element denoted by 0

(ii) $(F^*,\cdot)$ is an abelian group, where $F^* = \{F - \{0\}\}$. It's identity element is denoted by 1.

(iii) $\forall\ a,b,c \in F \qquad a.(b+c) = ab + ac$

* lemma 3.2 $F_p = \mathbb{Z}/p\mathbb{Z}$ where $p \in \mathbb{Z}$ is prime is an field.

Hint: Proof by contradiction. Let $p = a.b$ with $1 < a,b < n$

show that closure fails for F_p^*

* lemma 3.3 F_p^* is an group of order $p-1$ and is cyclic, ie

$$F_p^* = \{1, \alpha, \alpha^2 \ldots, \alpha^{p-2}\} = \{\bar{1}, \bar{2}, \ldots, \overline{p-1}\}$$

The generator α of F_p^* is called primitive root modulo p.

* Defn 3.4 The <u>characteristic of a field</u> is the order of $\langle 1 \rangle \subset (F,+)$

* lemma 3.5 <u>characteristic of a field</u>

- Finite → It is a prime number
- Infinite → Then it is called zero

Hint: Same as 3.2

* Defn 3.6 A vectorspace V over a field F is a set together two laws of composition, ie. vector addition and scalar multiplication s.t

(i) V is an abelian group with scalar addition. It's identity element is denoted by 0_V

(ii) if $a \in F$, $v \in V$ then $a.v \in V$

(iii) if $v \in V$ $\quad 1.v = v$

(iv) if $v, w \in V$ and $a, b \in F \longrightarrow$

$$a(v+w) = av + aw$$

$$(a+b)v = av + bv$$

$$a.(b.v) = (a.b)v$$

* Lemma 3.7 Let $v \in V$, $0_F \in F \implies 0_F.v = 0_V$

Linearly Independent Vectors

* Defn 3.8 A subspace of a vectorspace V over field F is a non empty subset s.t
① if $v, w \in V$ then $v + w \in V$
② if $a \in F$, $v \in V$ then $a.v \in V$

* Lemma 3.9 If W_1 and W_2 are subspaces of V then
① $W_1 \cap W_2$ is a subspace of V
② $W_1 + W_2$ is a subspace of V, where $W_1 + W_2$ is the group generated by $W_1 \cup W_2$ (2.25)

* Defn 3.10 Ordered Set → Order of elements matter
→ Repetition of elements is not allowed.
{Note: It is different from real analysis}

* Defn 3.11 The Span of an ordered set $S = \{v_1, v_2 \ldots\}$ with $v_i \in V$ is defined as the subspace generated by the linear combinations of v_i's, i.e.

$$\text{Span}(S) := \left\{ \sum_i a_i v_i \text{ where } a_i \in F, v_i \in V \right\}$$

* Defn 3.12 A vector space V is called finite dimensional if $\exists$ a finite set S s.t
$V = \text{Span}(S)$

* Defn 3.13 An ordered set $S = \{v_1, v_2 \ldots, v_n\}$ with $v_i \in V$ is called linearly independent if

$$a_1 v_1 + a_2 v_2 \ldots + a_n v_n = 0_V \implies a_i = 0 \text{ where } 1 \le i \le n$$

* Defn 3.14 (i) An empty set ϕ is linearly independent
(ii) $\text{Span}(\phi) = \{0_V\}$

* Defn 3.15 A basis $[B]$ of vectorspace V is an subset of the vectors of V s.t
① $[B]$ is linearly independent
② $\text{Span}(B) = V$

Direct Sum

* Lemma 3.16 Let L, S be subsets of finite dimensional V s.t L is linearly independent and $Span(S) = V$ then $\dim(Span(L)) \leq \dim(Span(S))$

Hint: Proof by contradiction $\dim(Span(L)) > \dim((Span S))$

* Defn 3.17 The dimension of a finite dimensional vectorspace V is defined as the number of vectors in its basis

* Lemma 3.18 Let V be a finite dimensional vectorspace and, $[\mathcal{B}]$ and $[\mathcal{B}]'$ are its two basis associated by $[\mathcal{B}]' = [\mathcal{B}]P \Longrightarrow$ P is invertible

P: base change matrix

If $v \in V$ then $v = [\mathcal{B}]'x' = [\mathcal{B}]x$ (x: coordinate vector of v w.r.t $[\mathcal{B}]$)

$x = Px'$

* Defn 3.19 The two subspaces U_1 and U_2 of V are called linearly independent if

$$U_1 \cap U_2 = \{0_v\}$$

* Defn 3.20 A vectorspace V is called direct sum of subspace $U_1, U_2 \dots, U_k$ if

① $U_1, U_2 \dots, U_k$ are linearly independent

② $V = U_1 + U_2 + \dots U_k$

It is denoted by $V = U_1 \oplus U_2 \oplus \dots \oplus U_k$

* Lemma 3.21 Let W_1 and W_2 are subspace of V then

$$\dim(W_1) + \dim(W_2) = \dim(W_1 + W_2) + \dim(W_1 \cap W_2)$$

Linear Transformation

* Defn 4.1 Let V, W be vectorspaces over the same field F. Then a map $T: V \to W$ is called a linear transformation if

① $T(v_1 + v_2) = T(v_1) + T(v_2)$ $\quad \forall v_1, v_2 \in V$

② $T(av) = aT(v)$ $\quad \forall v \in V, \forall a \in F$

* Defn 4.2

Kernel of $T := \{ v \in V : T(v) = 0_W \}$

Image of $T := \{ T(v) : v \in V \}$

* Lemma 4.3 Kernel is a subspace of V

Image is a subspace of W

* Lemma 4.4 $\dim(\text{Kernel}) + \dim(\text{Image}) = \dim(V)$

nullity rank

Hint: $[B] := \{ \underbrace{a_1, a_2, \ldots, a_k}_{\text{Basis of Kernel}}, \underbrace{b_1, \ldots, b_m}_{\text{Basis of } V} \}$

Show $\{T(b_1), T(b_2), \ldots, T(b_m)\}$ is a basis of the Image

* Lemma 4.5 Let $T: V \to W$ be a linear transformation, and P a subspace of V

then $T(P)$ is an subspace of the Image

* Defn 4.6 Let $T: V \to W$ be a linear transformation, $[B], [C]$ be the basis of V and W respectively

Then the matrix of linear transformation T wrt $[B]$ and $[C]$ is defined as a matrix A s.t

$$T([B]) = [C]A$$

Lemma If x, y be the coordinate vectors of $v \in V$ and $T(v) \in W$ wrt $[B]$ and $[C]$ respectively

then $Y = Ax$

Lemma If $[B']$ and $[C']$ are another sets of basis of V and W s.t

$$[B'] = [B]P \qquad [C'] = [C]Q$$

then A', the matrix of linear transformation T wrt $[B']$ and $[C']$ is given by

$$A' = Q^{-1} A P$$

* Defn 4.7 A linear transformation $T: V \to W$ is called an **Isomorphism** if T is bijective as a map.

Example: A n dim vectorspace V over field F is isomorphic to the vectorspace F^n.

* Lemma 4.8 Let $A_{n\times n}$ be a square matrix then

$$AX = 0 \iff A \text{ is invertible}$$

if and only if $x = 0$

Proof (i) A is invertible $\Rightarrow$ Let $AX = 0 \Rightarrow A^{-1}AX = A^{-1}0 \Rightarrow x = 0$

(ii) Let $AX = 0$ only for $x = 0 \Rightarrow$ columns of A are linearly independent

$[A]$ is an basis of $\mathbb{R}^n$, $[I]$ is another basis of $\mathbb{R}^n \therefore \exists P$ s.t. $[A]\underbrace{P}_{\text{invertible}} = [I]$

* Lemma 4.9 Let $A_{m\times n}$ be a matrix and $E_{m\times m}$ be an invertible matrix, then

$$\dim(\text{ColSpace}(A)) = \dim(\text{Column Space}(EA))$$

Hint use 4.4, 4.5

* Lemma 4.10 Let $A_{m\times n}$ be a matrix then

$$\dim(\text{ColSpace}(A)) = \dim(\text{RowSpace}(A))$$

Hint Use 1.8, 4.9

* Defn 4.11 A **linear operator** is a linear transformation $T: V \to V$

- Invertible / T is bijective / kernel $= \{0_v\}$ / Image $= V$ → Transformation matrix is invertible
- Singular → Transformation matrix is non invertible

* Lemma 4.12 Let $T: V \to V$ be a linear operator, then the following conditions are equivalent

(i) $V = \text{Kernel}(T) \oplus \text{Image}(T)$

(ii) $\text{Kernel}(T) \cap \text{Image}(T) = \{0_v\}$

(iii) $\text{Kernel}(T) + \text{Image}(T) = V$

These don't mean T is invertible

Eg: $\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$

Hint: Use 4.4, 3.21

Eigen Vector

* Lemma 4.13 — Let $T:V \longrightarrow V$ be a linear operator, $[B],[B']$ be two different basis of V associated by $[B'] = [B]P$ and A, A' be the transformation matrix of T wrt $[B]$ and $[B']$ then

$$A' = P^{-1}AP,$$ further

A and A' are called <u>similar matrices</u> as they represent the same linear transformation

* Defn 4.14 — Let $T:V \longrightarrow V$ be a linear operator, then a subspace W of V is called <u>T invariant</u> if $T(W) \subset W$

* Lemma 4.15 — Let $T:V \longrightarrow V$ be an linear operator, W be an <u>T-invariant</u> subspace of V, $\{w_1, w_2 \ldots, w_k\}$ be an basis of W, $[B] := \{w_1, w_2 \ldots, w_k, v_1, v_2 \ldots, v_n\}$ be a basis of V, $f: V \longrightarrow V/W$ be a map defined by $f(v+W) := \bar{v}$

(i) $\bar{v}_1 + \bar{v}_2 = \overline{v_1 + v_2}$

(ii) $a\bar{v}_1 = \overline{av_1}$

} Similar to 2.23

(iii) V/W along with (i) and (ii) forms a vectorspace

(iv) $f: V \longrightarrow V/W$ is a linear transformation

(v) $[\bar{B}] := [\bar{v}_1, \bar{v}_2 \ldots, \bar{v}_n]$ forms a basis of V/W.

(vi) $\varphi: V/W \longrightarrow V/W$ given by $\varphi(\bar{v}) := f(T(v))$ is an linear operator

(vii) The transformation matrix of T wrt $[B]$ has the form $A = \begin{bmatrix} P_{k\times k} & Q_{k\times n} \\ O_{n\times k} & R_{n\times n} \end{bmatrix}$

(vii)(a) The transformation matrix of φ wrt $[\bar{B}]$ is $R_{n\times n}$

* Defn 4.16 Eigen vector — Let V be a vectorspace over field F, $T:V \longrightarrow V$ a linear operator. Then $v \in V$ is called an eigen vector of T with eigen value λ if (i) $v \neq 0_v$

(ii) $T(v) = \lambda v$ where $\lambda \in F$

* Lemma 4.17 — An eigen vector is an one dimensional T-invariant subspace.

* Lemma 4.18 — Let V be a vectorspace, $T:V \longrightarrow V$ and $U:V \longrightarrow V$ be two linear operators, then $T+U: V \longrightarrow V$ where $(T+U)(v) := T(v) + U(v)$ is also a linear operator

Schur's lemma

* Lemma 4.19 Let $v \in V$ be an eigen vector of $T: V \to V$ with eigen value λ, then

$v \neq 0$, is in the null space of $\lambda I - T$ $\Leftarrow$ $\lambda I - T$ is singular $\Leftarrow$ $\{ \det(\lambda I - A) = 0$ (characteristic polynomial of T)

$\Downarrow$

Eigen values are roots of the characteristic polynomial

* Lemma 4.20
(i) The characteristic polynomial of $T: V \to V$ is independent of the choice of basis
(ii) Similar matrices have the same characteristic polynomial
(iii) The co-efficients of characteristic polynomial is independent of the choice of basis

* Lemma 4.21 Eigen vectors corresponding to distinct eigen values are linearly independent.

Hint: Proof by induction.

* Lemma 4.22 If $T: V \to V$ (n dim) has n distinct eigen values then $\exists$ basis $[\mathcal{B}]$ of V s.t the transformation matrix of T w.r.t $[\mathcal{B}]$ has the diagonal form, i.e. $\begin{bmatrix} \lambda_1 & & 0 \\ & \lambda_2 & \\ 0 & & \lambda_n \end{bmatrix}$

* Schur's Lemma 4.23 Let $T: V \to V$ (n dim V.S over F) be a linear operator,

If the characteristic polynomial of T factors into linear terms $\Longrightarrow$ $\exists$ a basis $[\mathcal{B}]$ of V s.t the transformation matrix of T w.r.t $[\mathcal{B}]$ is upper diagonal $\begin{bmatrix} R & \textcircled{a} \\ 0 & R \end{bmatrix}$

Hint Use 4.20(iv), 4.17, 4.15 (vi), (viii) to prove upper diagonal

Proof that the transformation is not always diagonalizable :-

Let v_1 be the eigen vector of $T: V \to V$, $V_1 := \text{Span}(v_1)$,

$f: V \to V/V_1$ where $f(v_2 v_1) = \bar{v}$

Let $\bar{v}_2$ be the eigen vector of $\varphi: V/V_1 \to V/V_1$ $\varphi(\bar{v}) := f(T(v))$

$v_2' = v_2 + c v_1$ $\quad T(v_2) = \lambda_2 v_2 + n v_1$

$T(v_2') = T(v_2) + T(c v_1) = \lambda_2 (v_2' - c v_1) + n v_1 + c \lambda_1 v_1 = \lambda_2 v_2' + (n + c(\lambda_1 - \lambda_2)) v_1$

then if $\lambda_1 \neq \lambda_2$ we can diagonalize

if $n = 0$ we can diagonalize

if $\lambda_1 = \lambda_2$ and $n \neq 0$ we cannot diagonalize

Generalized Eigen Vector

* Defn 4.24

Let V be a vectorspace over field F and $T: V \to V$ a linear operator. Then $x \in V$ is called a <u>generalized eigen vector</u> of T with eigen value λ and exponent d if

① $x \neq 0_V$ ② $(T-\lambda I)^d x = 0$

③ If $(T-\lambda I)^k x = 0 \implies k \geq d$

* Lemma 4.25

Let $x \in V$ be a generalized eigen vector of $T: V \to V$ with eigen value λ and exponent d,

$$u_i := (T-\lambda I)^i x \quad \text{with} \quad 0 \leq i \leq d-1$$

$$[\mathcal{B}] := \{u_1, u_2, \ldots, u_{d-1}\}$$

$$\mathcal{X} := \text{Span}([\mathcal{B}])$$

then

(i) $\mathcal{B}$ is linearly independent

(ii) $\mathcal{X}$ is T invariant

(iii) $T([\mathcal{B}])$ has the form $\begin{bmatrix} \lambda & & 0 \\ 1 & \lambda & \\ & \ddots & \ddots \\ 0 & & 1 \; \lambda \end{bmatrix}_{d \times d}$

Proof Hint (i) Show that any non trivial combination $[\mathcal{B}]$ is an genenerelized eigenvector $\therefore \neq 0_V$

* Lemma 4.26

A generalized eigen vector x of $T: V \to V$ has unique eigen value λ and exponent d

Proof Hint: Use binomial expansion

$$(x+y)^n = \sum_{k=0}^{n} {}^nC_k \, x^k y^{n-k}$$

* Jordan Decomposition Theorem 4.26

Let T be a linear operator on a finite dimensional complex vectorspace V. Then $\exists$ a basis $[\mathcal{B}]$ of V s.t the transformation matrix of T w.r.t $[\mathcal{B}]$ has the Jordan form.

Proof: We will prove ①-③. They together lead to the basis $[\mathcal{B}]$

① $V = \widehat{V_1}^{T\text{-invariant}} \oplus V_2 \oplus \cdots \oplus V_r$

② $V_i = \text{Span}([u_{i,0}, u_{i,1}, \ldots, u_{i,d_i-1}])$, $1 \leq i \leq r$

③ $u_{i,j} = (T-\lambda_i) v_{i,j-1}$, $1 \leq j \leq d_i - 1$

generalized eigen vector with eigen value λ_i and index d_i

Jordan Decomposition Theorem

We now that if V_i is 1 dim then $v \in V_i$ gives $T_{|V_i} : V_i \to V_i$ the jordan form.

We will use principle of induction, ie assuming the statement is true for complex vector space of dimensions 1 to $n-1$ prove it for n dim complex vectorspace.

* Let $T: V \to V$ be a linear operator on n dim complex vector space V. Since its characteristic polynomial factors into linear terms $\Longrightarrow$ $\exists$ an eigen vector v with eigen value λ.

If w.r.t basis $[\mathcal{B}]$ T has jordan form then so does $T-\lambda I$. Therefore for simplicity we we work with $T+\lambda I$ and refer to it as T, then

$T: V \to V$ is an linear operator and its nullspace is non-trivial.

Let K_i be the kernel and U_i be the image of $T^i : V \to V$. Since V is finite dimensional let $m \in \mathbb{N}$ be the smallest number s.t

$$K_i = K_j \quad \text{and} \quad U_j = U_j \qquad \forall\, i, j \geq m$$

then let $v \in K_m \cap U_m$, since $T^m v = 0$ and $\exists\, x \in V$ s.t $T^m x = v$

$T^{2m} x = 0 \Rightarrow x \in K^m \;\therefore\; v = T^m x = 0$

Using 4.12 $V = K_m \oplus V_m$

$\dim(V_m) < n$ $\therefore$ $\exists$ jordan generators. If $\dim(K_m) < n$ the proof is complete otherwise T is an idempotent operator.

* $T: V \to V$ be an idempotent eigen vector, prove $\exists$ a basis $[\mathcal{B}]$ s.t T has jordan form.

Let K be the kernel of T and U be the image of T. Since $\dim(U) < n$ $\exists$ jordan generators for it ie. $U = U_1 \oplus U_2 \cdots \oplus U_r$ and

$$U_i = \text{Span}([u_{i,0}, u_{i,1} \cdots u_{i,d_i-1}]), \qquad 1 \leq i \leq r,$$

$$u_{i,j} = (T)\, v_i, \qquad 1 \leq j \leq d_i - 1$$

generalized eigen vector with eigen value 0 and index d_i

Let $w_i \in V$ s.t $T w_i = v_i$ then w_i is an generalized eigenvector with eigen value 0 and index $d_i + 1$

Let $W_i = \text{Span}([u_{i,0}, u_{i,1} \cdots u_{i,d_i}])$ where $u_{i,j} = T^j w_i$

$W = W_1 + W_2 + \cdots + W_r$

Let $w \in W_n \cap W_k$ then $Tw \in V_n \cap V_k$ since $V_n \cap V_k = 0$

$w \in K$ Since $W_n \cap K = \text{Span}(u_{n,d_n}) = \text{Span}(v_{n,d_n-1})$

and $W_k \cap K = \text{Span}(u_{k,d_k}) = \text{Span}(v_{k,d_k-1})$

since v_{n,d_n-1} and v_{k,d_k-1} are linearly independent

$w = 0_v$

$\therefore W = W_1 \oplus W_2 \oplus \cdots \oplus W_r$

Let P = subspace of K that is not in W

Augment basis of P to W to get $[B]$ s.t T has jordan form w.r.t it.

* Lemma 4.27 Let J be a jordan block with eigenvalue λ then

(i) $J_\lambda = \lambda I + N$ where $N = \begin{bmatrix} 0 & & & \\ 1 & 0 & & \\ & \ddots & \ddots & \\ & & 1 & 0 \end{bmatrix}$

(ii) $\lambda I N = N \lambda I$

* Lemma 4.28 If T is a linear operator on a finite dimensional complex vectorspace V.

If for some $n \in \mathbb{Z}$, $T^n = I \implies T$ is diagonalizable

Orthogonal Linear Operator

* **Defn 5.1** $A \in \mathbb{R}^{n\times n}$ is called orthogonal if $A^TA = I$

* **Dot product** **Defn 5.2** Let $u, v \in \mathbb{R}^n$, $\underbrace{\langle u, v\rangle}_{\text{Dot Product}} := X^TY = x_1y_1 + x_2y_2 + \cdots + x_ny_n$ where $X = \{x_1, x_2, \ldots, x_n\}^t$, $Y = \{y_1, y_2, \ldots, y_n\}^t$ are the coordinate vectors of u, v wrt the standard basis

* **Lemma 5.3** $\forall\ u, v, w \in \mathbb{R}^n$ and $a \in \mathbb{R}$

(i) $\langle u, v+w\rangle = \langle u, v\rangle + \langle u, w\rangle$

(ii) $\langle u, av\rangle = a\langle u, v\rangle$

(iii) $\langle u, v\rangle = \langle v, u\rangle$

* **Defn 5.4** Let $v \in \mathbb{R}^n$, then $\underbrace{\|v\|}_{\text{length of } v} := \langle v, v\rangle$

* **Defn 5.5** $u, v \in \mathbb{R}^n$ are called orthogonal if $\langle u, v\rangle = 0$

* **Defn 5.6** An basis $[\mathcal{B}] = \{v_1, v_2, \ldots, v_n\}$ on $\mathbb{R}^n$ is called orthonormal if

$$\forall\ 1 \le i, j \le n \qquad \langle v_i, v_j\rangle = \begin{cases} 0 & \text{if } i \neq j \\ 1 & \text{if } i = j \end{cases}$$

* **Defn 5.7** A $\underbrace{T: \mathbb{R}^n \to \mathbb{R}^n}_{\text{linear operator}}$ is called orthogonal if $\forall\ u, v \in \mathbb{R}^n \qquad \langle T(u), T(v)\rangle = \langle u, v\rangle$

* **Lemma 5.8** An orthogonal linear operator $T: V \to V$ preserves the length of vectors

* **Lemma 5.9** The transformation matrix A of an orthogonal linear operator $T: \mathbb{R}^n \to \mathbb{R}^n$ w.r.t the standard basis is orthogonal, ie $A^TA = I$

* **Defn 5.10** A linear operator $T: \mathbb{R}^2 \to \mathbb{R}^2$ is called a reflection if it has two eigen vector v_1 and v_2 s.t (i) $\langle v_1, v_2\rangle = 0$ (ii) their eigen values are 1 and -1 respectively

* **Defn 5.11** $R_\theta = \begin{bmatrix} c\theta & -s\theta \\ s\theta & c\theta \end{bmatrix}$ $\quad S_0 = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$ $\quad R_\theta S_0 = S_0 R_{-\theta} = \begin{bmatrix} c\theta & s\theta \\ s\theta & -c\theta \end{bmatrix}$

* **Lemma 5.12** Let A be the transformation matrix of an orthogonal linear operator $T: \mathbb{R}^2 \to \mathbb{R}^2$ then

$A \in O_2$

$\det(A) = 1$ ↙ ↘ $\det(A) = -1$

$A = R_\theta$ ↓ Rotation about origin by θ

$A = R_\theta S_0 = S_0 R_{-\theta}$ ↓ Reflection about a line making $\theta/2$ w.r.t e_1

Hint:- $A = \begin{bmatrix} c\theta & * \\ s\theta & * \end{bmatrix}$ $\quad R^T A = \begin{bmatrix} 1 & 0 \\ 0 & \pm 1 \end{bmatrix}$

* **Defn 5.13** An orthogonal linear operator $T: \mathbb{R}^3 \to \mathbb{R}^3$ is called a rotation if

(i) It fixes a unit vector u, called it's pole

(ii) It rotates the 2D subspace othogonal to u

* **Lemma 5.14** If $A \in SO_3 \implies 1$ is an eigen value

Hint Prove $\det(A-I) = -\det(A-I)$
Use $\det(A) = \det(A^T) = 1$ and $AA^T = I$

* **Theorem 5.15** $\{$ Linear operators that act as rotation on $\mathbb{R}^3\}$ $\xrightarrow[\text{bijective}]{f}$ SO_3

* **Lemma 5.16** Let $M \in SO_3$ represents the rotation $\rho(u, \alpha)$

(i) Trace $(M) = 1 + \cos(2\alpha)$

(ii) If $B \in SO_3$ then $M' = BMB^T$ represents the rotation $\rho(u', \alpha)$ where $u' = Bu$.

Hint: (i) 4.20

(ii) to prove $\alpha' = \alpha$ use $f: \overline{B} \to \alpha'$ (rotation angle of $B \in SO_3$ → rotation angle of M')

intermediate Value theorem is continuous +

Cayley Hamilton Theorem

* Lemma 5.17 Let $\{p_k\}_{k=1}^{\infty}$ be a sequence of monic polynomial $p_k(t)$ of degree n with roots $\alpha_{k,1}, \alpha_{k,2}, \dots, \alpha_{k,n}$. Let $p(t)$ be another monic polynomial of degree n with roots $\alpha_1, \alpha_2, \dots, \alpha_n$ then

(i) If $\lim_{k\to\infty} \alpha_{k,i} = \alpha_i \quad \forall\ 1 \le i \le n$ then $\lim_{k\to\infty} p_k(t) \to p(t)$

(ii) If $\lim_{k\to\infty} p_{k_i}(t) = p(t)$ then the roots of each $p_k(t)$, ie. $\alpha_{k,i}$'s can be renumbered s.t

$$\lim_{k\to\infty} \alpha_{k,i} = \alpha_i \quad \forall\ 1 \le i \le n$$

Hint (i) co-efficients of $p_k(t)$'s are continuous functions of roots

(ii) Let's renumber different roots of $p_k(t)$'s s.t $|\alpha_{k,1} - \alpha_1|$ has the smallest magnitude then

$$|\alpha_1 - \alpha_{k,1}|^n \le |\alpha_1 - \alpha_{k,1}|\,|\alpha_1 - \alpha_{k,2}| \cdots |\alpha_1 - \alpha_{k,n}| = |p_k(\alpha_1)|$$

$$\lim_{k\to\infty} \alpha_{k,1} = \alpha_1$$

$p(t) = (t-\alpha_1) q(t)$ $\quad$ $p_k(t) = (t-\alpha_{k,1}) q_k(t)$ $\quad$ show $\lim_{k\to\infty} q_k(t) = q(t)$ and use induction

* Lemma 5.18 Let A be an $n \times n$ complex matrix then $\quad$ Hint: Use Schur's lemma 4.23

(i) $\exists\ \{A_k\}_{k=1}^{\infty}$ s.t $A_k \in \mathbb{C}^{n\times n}$, has distinct roots and $\lim_{k\to\infty} A_k = A$

(ii) If $\{A_k\}_{k=1}^{\infty}$ s.t $\lim_{k\to\infty} A_k = A \implies \lim_{k\to\infty} p_k(t) = p(t)$ where $p_k(t)$ and $p(t)$ are characteristic polynomial of A_k and A respectively

* Cayley Hamilton Theorem 5.19: Let $p(t) = t^n + c_{n-1}t^{n-1} \cdots + c_0$ be the characteristic polynomial of $A \in \mathbb{C}^{n\times n}$ then

$$p(A) = A^n + c_{n-1}A^{n-1} \cdots + c_0 I = 0_{n\times n}$$

Hint: 5.18, 4.21

* Lemma 5.20 Let $\sum_{k=1}^{\infty} m_k$ be a series of +ve real numbers s.t $\sum_{k=1}^{\infty} m_k$ is convergent. If $\{u_k(t)\}_{k=1}^{\infty}$ is sequence of functions $u_k : [r,s] \to \mathbb{R}$ s.t $\|u_k(t)\|_{[r,s]} \le m_k$. Then $\sum_{k=1}^{\infty} u_k(t)$ converges uniformly to some $f : [r,s] \to \mathbb{R}$.

Proof: Let $|f_k(t)| := |\sum_{i=1}^{k} u_i|$ then $|f_k(x)| = |\sum_{j=1}^{k} u_j(x)| \le \sum_{k=1}^{\infty} m_k < M$

$\therefore$ $\{f_k\}_{k=1}^{\infty}$ is a sequence of bounded functions

Let $\varepsilon > 0$, the sequence $\{S_k\}_{k=1}^{\infty}$ where $S_k := \sum_{i=1}^{k} m_i$ is convergent and therefore cauchy

$\exists\ L \in \mathbb{N}$ s.t. $\forall p, q \geq L$ with $p \geq q$ $\quad |\sum_{i=q+1}^{p} m_i| < \varepsilon$

Consider $x \in [r,s]$ $\quad |f_p(x) - f_q(x)| = |\sum_{i=q+1}^{p} u_i(x)| \leq \sum_{i=q}^{p} m_i < \varepsilon$

$\|f_p - f_q\|_{[r,s]} < \varepsilon$

$\therefore \{f_k(x)\}_{k=1}^{\infty}$ converges uniformly to $f: [r,s] \to \mathbb{R}$

* Lemma 5.21 Let $\{u_k(t)\}_{k=1}^{\infty}$ be a sequence of continuously differentiable functions on interval $[r,s]$. Suppose $\sum_{k=1}^{\infty} u_k(t)$ converges to $f(t)$ and $\sum_{k=1}^{\infty} u_k'(t)$ converges uniformly to $g(t)$. Then $\sum_{k=1}^{\infty} u_k(t)$ converges uniformly to $f(t)$, $f(t)$ is continuously differentiable and

$f'(t) = g(t)$

Hint :- Let $f_k(t) := \sum_{i=1}^{k} u_i(t)$ $\qquad f_k'(t) = \sum_{i=1}^{k} u_i'(t)$

Then use Theorem 7.9 from real analysis

* Defn 5.22 Vector valued function, matrix valued function and their derivatives

$$x(t) := \begin{bmatrix} x_1(t) \\ x_2(t) \\ \vdots \\ x_n(t) \end{bmatrix} \qquad x'(t) := \begin{bmatrix} x_1'(t) \\ x_2'(t) \\ \vdots \\ x_n'(t) \end{bmatrix} \qquad A(t) := \begin{bmatrix} a_{11}(t) & a_{12}(t) & \cdots & a_{1m}(t) \\ & a_{22}(t) & & \vdots \\ a_{n1}(t) & & \ddots & a_{nm}(t) \end{bmatrix} \qquad A'(t) := \begin{bmatrix} a_{11}'(t) & \cdots & a_{1m}'(t) \\ \vdots & \ddots & \vdots \\ a_{n1}'(t) & \cdots & a_{nm}'(t) \end{bmatrix}$$

* Lemma 5.23 If $A_1, A_2, \ldots, A_k$ are differentiable $n \times n$ matrices then

$$\frac{d}{dt}(A_1 A_2 \cdots A_k) = \sum_{i=1}^{k} A_1 A_2 \cdots A_{i-1} \frac{dA_i}{dt} A_{i+1} \cdots A_k$$

* Defn 5.24 If $a \in \mathbb{C}$ and $a = x + yi$ then $e^a = e^{x+yi} := e^x(\cos(y) + i \sin(y))$

* Lemma 5.25 (i) If $A, B \in \mathbb{C}^{n \times n}$ then $\|AB\| \leq n \|A\| . \|B\|$ where $\|A\|$ denotes maximum absolute value of A

(ii) If $A \in \mathbb{C}^{n \times n}$ then $\forall\ k > 0$ $\quad \|A^k\| \leq n^{k-1} \|A\|$

* Defn 5.26 If $A \in \mathbb{C}^{n \times n}$

$$e^A := I + \frac{A}{1!} + \frac{A^2}{2!} + \frac{A^3}{3!} + \cdots$$

Theorem 5.27 (i) The series e^A converges absolutely and uniformly for bounded sets of complex matrices.

(ii) e^{tA} is differentiable function of t and $\frac{d}{dt}(e^{tA}) = Ae^{tA}$

(iii) If $A, B \in \mathbb{C}^{n\times n}$ s.t $AB = BA$ then $e^{(A+B)} = e^A e^B$

Hint :- (i) Use 5.26, 5.25 and 5.20 to compute $|(e^A)_{ij}|$

(ii) Use 5.27(i) with 5.21 to compute $\frac{d}{dt}(e^{tA+sB})$

* **Lemma 5.28** e^A where $A \in \mathbb{R}^{n\times n}$ is an invertible matrix

* **Lemma 5.29** If A is a jordan block of size k : $e^A = e^{\lambda I + N} = e^{\lambda I} e^N$ (commute), compute the k-1 terms as $N^k = 0$

* **Defn 5.30** Dependent variable and its derivatives only appear raised to power 1 and not multiplied together; The highest derivative is first order

Linear, First Order, Constant co-efficient differential equation

$$\frac{dx}{dt} + P(y)x = Q(y) \longrightarrow (a)$$

If $Q(y) = 0$ then the equation is called homogeneous linear, first order, constant co-efficient differentiable equation.

$$\frac{dx}{dt} + P(y)x = 0 \longrightarrow (b)$$

* Every solution of (a) is called a particular solution
* Solutions of (b) are called Homogeneous solution

* **Lemma 5.31** For equation (a): General Solution = Particular solution + Homogeneous Solution

Proof :- Let $x_1(t)$ and $x_2(t)$ be two particular solutions then

$$\frac{d}{dt}(x_2(t) - x_1(t)) + P(y)(x_2(t) - x_1(t)) = Q(y) - Q(y) = 0$$

$x_2(t) - x_1(t) = x_h(t)$ } homogeneous solution

$x_2(t) = x_1(t) + x_h(t)$

* Lemma 5.32

Let $x(t)$ be a complex valued solution of $\frac{dx}{dt} = \underset{n\times n}{A}\, x$, A real matrix $\Longrightarrow$ The real and complex part of $x(t)$ solve the same equation

Proof Let $x(t) = u(t) + iv(t)$

$$\frac{d}{dt}(u+iv) = Au + iAv$$

Comparing real and complex parts

$$\frac{du}{dt} = Au \qquad \frac{dv}{dt} = Av$$

* Lemma 5.33

Let $A \in \mathbb{R}^{n\times n}$ and let P be an invertible matrix s.t $A' = P^{-1}AP = \begin{bmatrix} \lambda_1 & & 0 \\ & \lambda_2 & \\ 0 & & \lambda_n \end{bmatrix}$

Then the general solution of $\frac{dx}{dt} = Ax$ is $x = P\tilde{x}$ where

$\tilde{x} = (c_1 e^{\lambda_1 t}, c_2 e^{\lambda_2 t}, \dots, c_n e^{\lambda_n t})$ and solves $\frac{d\tilde{x}}{dt} = A'\tilde{x}$

* Theorem 5.34

Let $A \in \mathbb{R}^{n\times n}$ or $\mathbb{C}^{n\times n}$. The columns of matrix e^{tA} form a basis for the solution space of $\frac{dx}{dt} = Ax$

Proof $\frac{d}{dt}\left(e^{-tA}x(t)\right) = -Ae^{-tA}x(t) + e^{-tA}Ax = 0$

$\therefore e^{-tA}x(t) = c$

$x(t) = e^{tA}c$

Symmetry

* **Defn 6.1** Distance b/w $P, Q \in \mathbb{R}^n$

$$|\vec{PQ}| := |v - u|$$

(v: vector pointing to Q from origin; u: vector pointing to P from origin; $|v-u|$: length of vector 5.4)

* **Defn 6.2** Isometry (Distance preserving map):

$f: \mathbb{R}^n \to \mathbb{R}^n$ s.t $\forall\ u, v \in \mathbb{R}^n$

$$|f(u) - f(v)| = |u - v|$$

* **Lemma 6.3** Let $T: \mathbb{R}^n \to \mathbb{R}^n$ be a map, then

T is an isometry that fixes the origin $\iff$ T is an orthogonal linear operator

Hint: if $\langle x, y\rangle = \langle x, x\rangle = \langle y, y\rangle$ then $x = y$

* **Lemma 6.4** All the isometries of $\mathbb{R}^n$ together form a group denoted by M_n

* **Lemma 6.5** If $m \in M_n$, then $\exists$ a unique translation $t_a \in M_n$ and an orthogonal linear operator $\varphi \in M_n$ s.t

$$m = t_a \varphi$$

Hint 6.3

* **Lemma 6.6** Let t_a, φ be a translation and a orthogonal linear operator of $\mathbb{R}^n$ respectively then

$$\varphi t_a = t_{\varphi(a)} \varphi$$

* **Lemma 6.7** (i) $\pi: M_n \to O_n$ defined by $\pi(t_a \varphi) := \varphi$ is a group homomorphism

(ii) $m \in M_n$: $\det(\pi(m)) = 1$ → Orientation preserving isometry; $\det(\pi(m)) = -1$ → Orientation reversing isometry

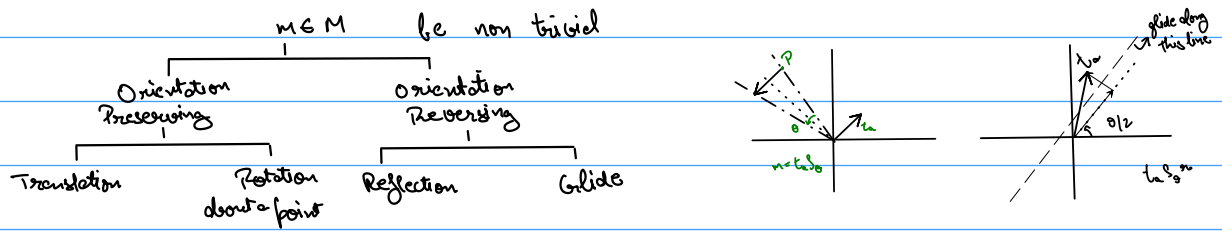
* **Lemma 6.8** Let C_1, C_2 be two coordinate systems (Origin + Orthonormal Basis) of $\mathbb{R}^n$ associated with an isometry $\eta \in M_n$, ($P_{C_1} = \eta P_{C_2}$) $T \in M_n$ s.t its formula w.r.t C_1 and C_2 are m_1 and m_2 respectively then

$$m_2 = \eta^{-1} m_1 \eta$$

* **Defn 6.9** M: Isometry group of $\mathbb{R}^2$.

↳ generated by: t_a (Translation by vector a), S_θ (rotation by angle θ about origin), r (reflection about e_1 axis)

* Lemma 6.10



* Lemma 6.11 If $G \subset M$ s.t it preserves a point $p \Longrightarrow$ $G \subset t_p O_\theta t_p^{-1}$

* Lemma 6.11.1 If $G \subset M$ contains rotations about two different centers $\Longrightarrow$ G contains a pure translation

Hint: let $h_1, h_2 \in G$ be the two rotations then $h_3 = h_1 h_2 h_1^{-1} h_2^{-1}$ is a pure translation

* Defn 6.12 $\Gamma \in (\mathbb{R}, +)$ is called discrete if $\exists\, \varepsilon > 0$ s.t $\overbrace{|a|}^{\text{absolute value}} > \varepsilon \quad \forall a \in \Gamma$

* Lemma 6.13 If $\Gamma \in (\mathbb{R}, +)$ is discrete $\Longrightarrow$ $\Gamma = a\mathbb{Z}$ where $a \in \mathbb{R}$

* Defn 6.14 $G \subset M$ is called discrete if $\exists \{m_k\}_{k=1}^{\infty} \subset G$ s.t then $\lim_{k\to\infty} m_k = I$

* Lemma 6.15 $G \subset M$ is discrete $\Longleftrightarrow$ $\exists\, \varepsilon > 0$ s.t $\rightarrow$ $\forall\, t_a \in G$ with $a \neq 0$ $\overbrace{|a|}^{\text{length of vector}} > \varepsilon$
$\rightarrow$ $\forall\, t_a S_\theta \in G$, $\theta \neq 0$ then $|\theta| > \varepsilon$

Hint: $A \Leftarrow B$, Proof by contrapositive. $\exists \{m_k\}_{k=1}^{\infty} \subset G$ $\lim_{k\to\infty} m_k = I$ and $\forall j, k \in \mathbb{N}$ $m_j \neq m_k$ if $j \neq k$ then

Case 1 $\{m_k\}_{k=1}^{\infty}$ has infinite translation. $\lim_{k\to\infty} m_k = I \rightarrow \lim_{i\to\infty} t_{a_{k_i}} = 0_V$.

Case 2 $\{m_k\}_{k=1}^{\infty}$ has infinite orientation reversing elements. Det is continuous function of matrix elements.

Case 3 $\{m_k\}_{k=1}^{\infty}$ has infinite rotations. $\lim_{k\to\infty} m_k = I \Longrightarrow \lim_{i\to\infty} t_{a_{k_i}} = 0_V$ and $\lim_{i\to\infty} S_{\theta_{k_i}} = I$

$A \Rightarrow B$, Proof by contrapositive. Goal construct $\{m_k\}_{k=1}^{\infty} \subset G$ $\lim_{k\to\infty} m_k = I$ and $\forall j, k \in \mathbb{N}$ $m_j \neq m_k$ if $j \neq k$

Case 1 $\nexists\, \varepsilon > 0$ for $t_a \in G$

Case 2 $\nexists\, \varepsilon > 0$ for $\overbrace{t_a S_\theta}^{\text{hx}} \in G \Longrightarrow G$ has infinite rotations

- a) Let $t_b \in G$ then $h_c = h_k t_b h_k^{-1} t_b^{-1}$ that depends on $|b|$ and $\theta/2$
- b) Rotations about multiple centers. Use 6.11.1
- c) Rotations about single centers.

* **Lemma 6.16** Let $G \subset O_2$, then G has finite order $\iff$ G is discrete

  - C_n: Cyclic group generated by s_θ where $\theta = \frac{2\pi}{n}$ and $n \in \mathbb{N}$
  - D_n: Dihedral group if coordinates system is selected appropriately generated by r and s_θ and $\theta = \frac{2\pi}{n}$ and $n \in \mathbb{N}$

* **Fixed Point Theorems 6.17** If $G \subset M$ has finite order $\Rightarrow$ G fixes a point and $G = C_n$ or D_n if coordinates are selected appropriately.

* **Defn 6.18** The translation group $G \subset M$ is defined as $L := \{ v \in \mathbb{R}^2 \text{ s.t } t_v \in G \}$

* **Lemma 6.19** Let $L \subset (\mathbb{R}^2, +)$ be discrete then
  (i) A bounded region of $(\mathbb{R}^2, +)$ contains only finitely many point of L
  (ii) If L is non trivial then it contains a non zero vector of minimal length.

* **Lemma 6.20** Let $\mathcal{B} = \{u, v\}$ be a basis of $(\mathbb{R}^2, +)$, $L := \{ mu + nv \text{ where } m, n \in \mathbb{Z} \}$
  $\pi(\mathcal{B}) := \{ p \in \text{interior of parallelogram formed by vertices } \{0, u, v, u+v\} \text{ or along the edges } [0,u) \text{ or } [0,v) \}$
  then $\forall w \in (\mathbb{R}^2, +)$ $\exists\, x \in L$ and $w_0 \in \pi(\mathcal{B})$ s.t $w = x + w_0$. Hint division by remainder

* **Lemma 6.21** If $L \subset (\mathbb{R}^2, +)$ is discrete then
  - $L = \{0\}$
  - $L = \{ a\mathbb{Z} \text{ where } a \in \mathbb{R}^2 \}$
  - $L = \{ a\mathbb{Z} + b\mathbb{Z} \text{ where } a, b \in \mathbb{R}^2 \}$

* **Defn 6.22** The point group $\bar{G}$ of $G \subset M$ is defined as $\bar{G} := \{ \varphi \in O_2 \text{ s.t } t_a \varphi \in G \}$

* **Theorem 6.23** Let $G \subset M$ be discrete, $a \in L$, $\bar{g} \in \bar{G}$ and $a' = \bar{g}(a) \Rightarrow a' \in L$ Hint : Use conjugation

* **Lemma 6.24** Crystallographic restriction :- Let L be a non trivial discrete subgroup of $(\mathbb{R}^2, +)$, $H \subset O_2$ be the subgroup of symmetries of L, then
  - Every rotation in H has one of these order $1, 2, 3, 4, 6$
  - H has the structure of C_n or D_n with $n = 1, 2, 3, 4, \text{or } 6$

* Note 6.25 The point group $\bar{G}$ and the translation group L of a discrete group G doesn't identify it completely.

* Lemma 6.26 The concept of crystallographic restrictions limits the number of discrete subgroups of M to 17.

* Note 6.27 The concept of crystallographic restrictions can be extended to 3D to conclude there are 230 discrete groups in 3D.

* Defn 6.28 Let G be a group and S be a set. Then the map $f: G \times S \to S$ is called a group operation if

(i) $e.s = s \quad \forall s \in S$

(ii) $g_1.(g_2.s) = (g_1.g_2).s \quad \forall s \in S, \forall g_1, g_2 \in G$

* Defn 6.29 Let $f: G \times S \to S$ be a group operation, $s \in S$

Orbit of s $\quad O_s := \{ g.s : g \in G \}$

Stabilizer of s $\quad G_s := \{ g \in G : g.s = s \}$

* Defn 6.30 A group operation $f: G \times S \to S$ is called transitive if S consists of a single orbit under f.

* Example Let $H \subset G$, then $f: G \times G/H \to G/H$ defined by $f(g, \bar{a}) = \overline{ga}$ is a transitive group action

* Lemma 6.31 Let $f: G \times S \to S$ a group operation, $s \in S$, $H \subset S$ be the stabilizer of s then

(1) If $a, b \in G$ s.t $as = bs \implies a^{-1}b \in H$

(2) If $s' \in S$ s.t $s' = g.s$ then the stabilizer of s', ie $H' = gHg^{-1}$

* Theorem 6.32 Let $f: G \times S \to S$ a group operation, $s \in S$, O_s and $H \subset S$ be the orbit and stabilizer of s. Then $\exists$ a bijective map $\varepsilon: G/H \to O_s$ where $\varepsilon([aH]) := a.s$.

Also $\quad |G| = |H|\,|G/H|$ $\quad$ } Counting Formula

$\quad\quad = |H|\,|O_s|$

* Lemma 6.33 — The relation of belonging to an orbit is an equivalence relation $\Rightarrow$ The orbits partition the set S & $|S| = |O_{s,1}| + |O_{s_2}| + \cdots + |O_{s_k}|$

* Defn 6.34 — The permutation representation of $f: G \times S \to S$ a group operation is a group homomorphism $\varphi: G \to \text{Perm}(S)$
(φ: permutation representation; Perm(S): Group of permutations of S)

* Lemma 6.35 — $\{ f: G \times S \to S \}$ (Operation of G on set S) $\xleftrightarrow{\text{bijective correspondence}}$ $\{ \varphi: G \times S \to S \}$ (permutations representations)

* Defn 6.36 — A group operation $f: G \times S \to S$ is called faithful if the only element $g \in G$ s.t. $g.a = a$ $\forall a \in S$ is identity, i.e. $e \in G$.
Restatement: The kernel of the permutation representation is $\{e\}$

* Theorem 6.37 — Let $G \subset SO_3$ be a finite ordered subgroup. Let P be the set of poles of elements of G. Then $f: G \times P \to P$ is a valid group operation where
$f(g,p) := g(p)$

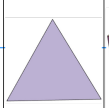
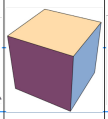
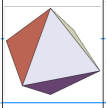
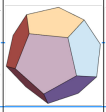
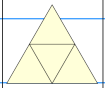
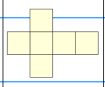
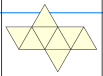
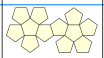
* Lemma 6.38 — A finite subgroup of SO_3 has one of the following structure
(i) C_k :- Cyclic of order k
(ii) D_k :- Dihedral group of order $2k$.
(iii) T :- Tetrahedral group of 12 rotational symmetries
(iv) O :- Octahedral group of 24 rotational symmetries
(v) I :- Icosahedral group of 60 rotational symmetries

Hint (a) Use 6.37 & count the number of fixes to get
$\sum_{O_p} \left(1 - \frac{1}{r_p}\right) = 2 - \frac{2}{N}$
(O_p: Orbit of pole p; $\#$ of elements with $p \in P$ as pole; N: Order of group)
Also $N = r_p \, n_p$ (n_p: # elements in the orbit of $p \in P$)

(b) Use $1 \leq 2 \cdot 2/N < 2$

(c) $1 + 2/N = 1/q_1 + 1/q_2 + 1/q_3$ ∴ It is not possible $q_1 > 3$, $q_2 > 3$ and $q_3 > 3$

* Lemma 6.39 There exist only 5 regular polyhedron

regular tetrahedron	cube	regular octahedron	regular dodecahedron	regular icosahedron
				
				

Symmetry Groups: T O O I I

p Groups

* **Defn 7.1** Let G be a group. Then $f: G \times G \to G$ defined by $f(g,x) := gxg^{-1}$ is called conjugation and is a valid group operation

* **Defn 7.2** Let G be a group. Then

(i) $Z := \{x \in G : x.g = g.x \ \forall g \in G\}$ is called center of G

(ii) $Z(x) := \{g \in G : g.x = xg\}$ is called Centralizer of x

* **Lemma 7.3** Let G be a group and $x \in G \implies Z \subset Z(x)$, $x \in Z(x)$

* **Lemma 7.4** (i) $|G| = |Z(x)| \, |C(x)|$ Hint 6.32

$C(x)$: Conjugacy class of x

(ii) $|G| = |C(x_1)| + |C(x_2)| - \cdots + |C(x_k)|$

* **Lemma 7.5** (i) If $x \in Z \implies C(x) = \{x\}$

(ii) If $x \in H$ and $H \trianglelefteq G \implies C(x) \subset H$

* **Defn 7.6** A group G is called p group if $|G| = p^e$ where p is a prime number and $e \geq 1$

* **Lemma 7.7** The center Z of a p group is non trivial. Hint 7.5 (ii)

* **Lemma 7.8** Let G be a p group, S be a set s.t $|S| = n$ and n is not divisible by p, $\mathcal{S}: G \times S \to S$ be a valid group operation, then $\exists\, s \in S$ s.t $|O_s| = 1$.

Hint 7.5 6.32

* **Lemma 7.9** Every group G of order p^2 where p is a prime number is :

(i) Abelian. Hint 7.4

(ii) Either cyclic or isomorphic to the product of two cyclic groups of order p.

* **Defn 7.10** A group G is called simple if it has no proper normal group

* **Defn 7.11** Let G be a group and $S := \{H \subset G\}$ (set of subgroups of G) then

$f: G \times S \to S$ defined by $f(g,H) := gHg^{-1}$ is a valid group operation

$\underbrace{N(H)}_{\text{stabilizer of } H} = \{g \in G : gHg^{-1} = H\}$

* **Lemma 7.12** Let G be a group and $H \subset G$ then

- $H \trianglelefteq N(H) \subset G$
- $H \trianglelefteq G \iff N(H) = G$

* **Lemma 7.13** Let S be a set s.t $|S| = p^e m$ where p is a prime number, $e \geq 1$, and m is not divisible by p

$U := \{A \subset S \text{ s.t } |A| = p^e\}$

then $|U| = {}^{p^e m}C_{p^e}$ is not divisible by p

* **Lemma 7.14** If G is a group, $S := \{A \subset G\}$ (set of subset of G) then

(i) $f: G \times S \to S$ where $f(g,A) := gA$ } is a valid group operation.

(ii) if $U \subseteq S$, $H \subset G$ is stabilizer of U then $|H|$ divides $|G|$ and $|U|$

Hint: $H \times U \to U$ (right cosets)

$|U| = |O_{u_1}| + |O_{u_2}| + \cdots + |O_{u_k}|$

* **Defn 7.15** Let G be a group s.t $|G| = p^e m$ where p is a prime number, $e \geq 1$, and m is not divisible by p.

$H \subset G$ is called sylow's p group if $|H| = p^e$.

* **Sylow's First Theorem 7.16** Let G be a group s.t $|G| = p^e m$ where p is a prime number, $e \geq 1$, and m is not divisible by p.

Then $\exists\, H \subset G$ s.t H is a sylow's p group

Hint $f: G \times S \to S$ (Set of subset of G of order p^e)

Use 7.13, 7.8

$|S| = |O_{s_1}| + |O_{s_2}| + \cdots + |O_{s_k}|$

* Second Sylow's theorem 7.17

Let G be a group s.t $|G| = p^e m$ where p is a prime number, $e \geq 1$, and m is not divisible by p

(i) If $K \subset G$ is a p subgroup and $\Rightarrow$ $\exists g \in G$ s.t $K \subset gHg^{-1}$
$H \subset G$ a sylow's p group

(ii) All sylow's p subgroups are conjugates of each other.

Hint $f: G \times G/H \to G/H$ where $f(g,[x]) = g[x]$ is a transitive group operation (using 6.30)
Stabilizer of $[H]$ is H itself.

consider $f': K \times G/H \to G/H$

then $|G/H| = m$ is not divisible by $|K|$ using 7.8

$\exists [L] \in G/H$ s.t $[L]$ is fixed by $K \Rightarrow K \subset M_{[L]}$ where $M_{[L]} \subset G$ is stabilizer of $[L]$

also $\exists g \in G$ s.t $g[H] = [L] \Rightarrow M_{[L]} = gHg^{-1}$ using 6.31

$K \subset gHg^{-1}$

* Sylow's Third Theorem 7.18

Let G be a group s.t $|G| = p^e m$ where p is a prime number, $e \geq 1$, and m is not divisible by p

If s is the number of sylow's p subgroup, then

(i) m is divisible by s

(ii) $s = kp+1$

Hint $f: G \times \overbrace{S}^{\text{set of subgroups } p \text{ subgroup}} \to S$ defined by $f(g,H) := gHg^{-1}$ is a transitive group operation (7.17)

let $H \in S$ then $|G| = |N(H)|\,|O_H|$

$= |N(H)|\,|S|$ and $H \subset N(H) \Rightarrow p^e \leq |N(H)| \leq p^e m$

$p^e m = p^e a s$ where $e \in \mathbb{N}$

$\therefore$ s divides m.

(ii) $f': H \times S \to S$ then $|O_H| = 1$

let $H' \in S$ s.t $|O_{H'}| = 1$ then $H \subset N(H') \subset G$

H', H are sylow's p subgroup of $N(H')$ where $|N(H')| = p^e a$ $\therefore$ $\exists g \in N(H')$

s.t $gH'g^{-1} = H$ since $H' \trianglelefteq N(H') \Rightarrow H = H'$ $\therefore$ only H has orbit 1

Bilinear Forms

* Defn 8.1 **Normal Matrix**

$A^*A = AA^*$

Symmetric Matrix	Hermitian Matrix	Orthogonal Matrix	Unitary Matrix
$A^T = A$	$A^* = A$	$A^TA = I$	$A^*A = I$

* Defn 8.2 **Symmetric Bilinear Form**

$\langle , \rangle : V \times V \longrightarrow \mathbb{R}$ s.t $\forall\ u, v, w \in V, a \in \mathbb{R}$

(i) $\langle u+w, v \rangle = \langle u, v \rangle + \langle w, v \rangle$

(ii) $\langle au, v \rangle = a \langle u, v \rangle$

(iii) $\langle u, v \rangle = \langle v, u \rangle$

Hermitian Form

$\langle , \rangle : V \times V \longrightarrow \mathbb{C}$ s.t $\forall\ u, v, w \in V, a \in \mathbb{C}$

(i) $\langle u+w, v \rangle = \langle u, v \rangle + \langle w, v \rangle$

(ii) $\langle au, v \rangle = a^* \langle u, v \rangle$

(iii) $\langle u, v \rangle = \overline{\langle v, u \rangle}$

* Defn 8.3 Let $\langle , \rangle$ be a bilinear form on a real vector space V or a Hermitian form on a complex vectorspace V and $[B] = \{v_1, v_2, \dots, v_n\}$ be a basis of V. Then the matrix of the form $\langle , \rangle$ w.r.t $[B]$ is defined as $A_{n \times n}$ where $a_{ij} := \langle v_i, v_j \rangle$

* Lemma 8.4 (i) The matrix of symmetric bilinear form over a real vector space is symmetric

(ii) " " " Hermitian " " , complex " " hermitian

* Lemma 8.5 Let $\langle , \rangle : V \times V \longrightarrow \mathbb{R}$ be a symmetric bilinear form, A its matrix w.r.t basis $[B]$ of V, $v, w \in V$ with coordinates vectors x, y w.r.t $[B]$ then

$$\langle v, w \rangle = x^T A y$$

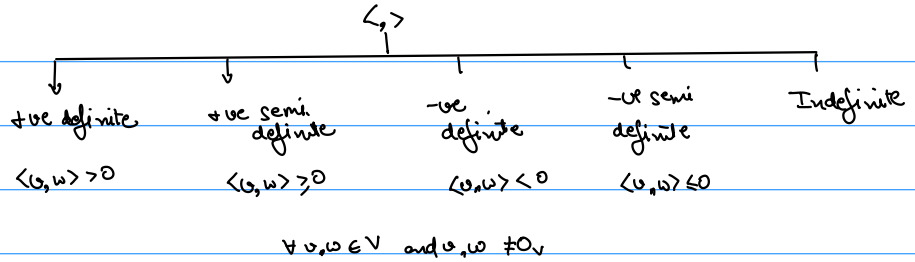
* Lemma 8.6 Let $\langle , \rangle : V \times V \longrightarrow \mathbb{C}$ be a Hermitian form, A its matrix w.r.t basis $[B]$ of V, $v, w \in V$ with coordinates vectors x, y w.r.t $[B]$ then

$$\langle v, w \rangle = x^* A y$$

* **Lemma 8.7** Let $\langle , \rangle : V \times V \to \mathbb{R}$ be a symmetric bilinear form, $[B], [B']$ be basis of V associated by $[B'] = [B]P$ and A, A' be matrices of $\langle , \rangle$ w.r.t basis $[B], [B'] \implies A' = P^T A P$

* **Lemma 8.8** Let $\langle , \rangle : V \times V \to \mathbb{C}$ be a Hermitian form, $[B], [B']$ be basis of V associated by $[B'] = [B]P$ and A, A' be matrices of $\langle , \rangle$ w.r.t basis $[B], [B'] \implies A' = \bar{P}^T A P$

* **Defn 8.9**



* **Caution** From here on upto lemma 8.23 unless mentioned assume the form can be symmetric bilinear or Hermitian.

* **Defn 8.10** Let $v, w \in V$, then v is called orthogonal to w w.r.t the form $\langle , \rangle$ ie. $v \perp w$ if $\langle v, w \rangle = 0$

* **Defn 8.11** Let $\underset{\text{subspace}}{W} \subset V$, then the orthogonal space of W ie $W^\perp := \{ u \in V \text{ s.t } \langle u, v \rangle = 0 \ \forall v \in V \}$

* **Defn 8.12** $v \in V$ is called null vector if $\langle u, v \rangle = 0 \quad \forall u \in V$

* **Defn 8.13** $\underset{\text{subspace of } V}{N} := \{ v \in V \text{ s.t } \langle v, u \rangle = 0 \ \forall u \in V \}$
Nullspace

* **Defn 8.14** $\langle , \rangle$ is called non degenerate on V if $N = \{0_V\}$

* **Defn 8.15** $\langle , \rangle$ is called non degenerate on $\underset{\text{subspace}}{W} \subset V$ if $w \in W$ and $w \neq 0_V \implies \exists v \in W$ s.t $\langle v, w \rangle \neq 0$

* **Lemma 8.16** If $\langle , \rangle$ is non degenerate on $W \subset V \iff W \cap W^\perp = \{0_V\}$

* Caution — A form may be non degenerate on V but degenerate on $W \subset V$ and vice versa

* Lemma 8.17 — Let $\langle , \rangle$ be a form on vectorspace V and A be its matrix
$\langle , \rangle$ is non degenerate $\iff$ A is invertible

* Theorem 8.18 — Let $\langle , \rangle$ be a form on vectorspace V, $W \subset V$ (subspace), and $W^\perp$ is orthogonal space of W, then
$\langle , \rangle$ is non degenerate on W $\iff$ $V = W \oplus W^\perp$ (direct sum)

Hint $A \Rightarrow B$ $\langle , \rangle$ non degenerate on $W \Rightarrow W \cap W^\perp = \{0_v\}$.
We just need to show $V = W + W^\perp$
Let $[\mathcal{B}] = \{ \underbrace{w_1, w_2, \dots, w_k}_{\text{Basis of } W}, v_1, v_2, \dots, v_{n-k} \}$ be an basis of V

Matrix of $\langle , \rangle$ wrt. $[\mathcal{B}]$ $M = \begin{bmatrix} A & B \\ C & D \end{bmatrix}$. Since $\langle , \rangle$ is non degenerate on W A is invertible
consider $[\mathcal{B}'] = [\mathcal{B}] P$ where $P := \begin{bmatrix} I & Q \\ 0 & I \end{bmatrix}$ then $M' = P^* M P = \begin{bmatrix} A & AQ+B \\ * & * \end{bmatrix}$
Since A is invertible $Q = -A^{-1}B$ which gives $M' = \begin{bmatrix} A & 0 \\ 0 & L \end{bmatrix}$

* Lemma 8.19 — If $\langle , \rangle$ a symmetric or Hermitian form on V is not identically zero $\Longrightarrow$ $\exists\, v \in V$ s.t $\langle v, v \rangle \neq 0$

Hint: $\exists\, u, v \in V$ s.t $\langle u, v \rangle \neq 0$ consider $\langle u+v, u+v \rangle$

cw 8.20 — Let $\langle , \rangle$ be a form on $V \Rightarrow$ $\exists$ a basis $[\mathcal{B}]$ of V s.t the elements of $[\mathcal{B}]$ are othogonal to each other.

Hint: Case 1 $\langle , \rangle$ is identically 0
Case 2 $\langle , \rangle$ is not identically 0 $\overset{8.9}{\Longrightarrow} \exists\, u \in V$ s.t $\langle u, u \rangle \neq 0 \overset{8.18}{\Longrightarrow} V = \overline{U} \oplus U^\perp$ (span of u)
Use induction on dimension of V

* Theorem 8.21 — Given $\langle , \rangle$ form on V, $\exists$ a basis $[\mathcal{B}]$ of V s.t the matrix of the form has the structure $A = \begin{bmatrix} I_m & & \\ & -I_n & \\ & & 0 \end{bmatrix}$

Hint Use 8.20

* Defn 8.22 Let $\langle , \rangle$ be non degenerate on $W \subset V$ $\overset{8.18}{\Longrightarrow}$ $V = W \oplus W^\perp$

Orthogonal Projection $\pi : V \to V$

if $v \in V$ $\quad \pi(v) := w$ where $w \in W$ and $w' \in W^\perp$ s.t $v = w + w'$

* π is a linear transformation

* Lemma 8.23 Let $\langle , \rangle$ be non degenerate on $W \subset V$, $[\mathcal{B}] = \{ \overbrace{w_1, w_2 \ldots, w_k}^{\text{Basis of } W}, \overbrace{w_{k+1} \ldots w_n}^{\text{Basis of } W^\perp} \}$ an orthogonal basis then

$$\pi(v) = c_1 w_1 + c_2 w_2 \ldots + c_k w_k$$

Projection Formula. where $c_i = \dfrac{\langle v, w_i \rangle}{\langle w_i, w_i \rangle}$

Hint: $\langle v, w_1 \rangle$

* Defn 8.24 Euclidean space : Real vectorspace V + $\langle , \rangle$ a + ve definite symmetric bilinear form

* Defn 8.25 Hermitian space : Complex vectorspace V + $\langle , \rangle$ a + ve definite Hermitian form

* Lemma 8.26 Let V be a euclidean or hermitian space, then

$W \subset V$ a subspace $\Longrightarrow$ $V = W \oplus W^\perp$ Hint 8.25, 8.18

* Defn 8.27 Let $T : V \to V$ be a linear operator on a Hermitian space and A be its transformation matrix w.r.t a orthonormal basis $[\mathcal{B}]$

(i) T is called normal operator if A is normal

(ii) T " " hermitian " " A " hermitian

(iii) T " " unitary " " A is unitary

(iv) Adjoint operator $T^* : V \to V$ is the linear operator whose matrix wrt $[\mathcal{B}]$ is A^*

* Lemma 8.28 Let $T : V \to V$ and $U : V \to V$ be linear operators on a hermitian space then

$$(T+U)^* = T^* + U^* \quad , \quad (TU)^* = U^* T^* \quad , \quad T^{**} = T$$

* Lemma 8.29 — Let A be a normal, hermitian or unitary matrix and P be a unitary matrix. Then $A' = P^* A P$ is the same i.e. normal, hermitian or unitary.

* Lemma 8.30 — Let $T: V \to V$ be a linear operator on a Hermitian space and T^* be the adjoint operator, then $\forall v, w \in V$

(i) $\langle Tv, w \rangle = \langle v, T^* w \rangle$ and $\langle v, Tw \rangle = \langle T^* v, w \rangle$

(ii) if T is normal $\langle Tv, Tw \rangle = \langle T^* v, T^* w \rangle$

(iii) " " " Hermitian $\langle Tv, w \rangle = \langle v, Tw \rangle$

(iv) " " " Unitary $\langle Tv, Tw \rangle = \langle v, w \rangle$

Hint: Hermitian space $\Rightarrow$ find a orthonormal basis $[\mathfrak{B}]$ of V s.t. Matrix of the form $\langle\,\rangle$ is I

* Lemma 8.31 — Let $T: V \to V$ be a linear operator on a Hermitian space V and $W \subset V$ (subspace).

W is T invariant $\iff$ $W^\perp$ is T^* invariant

Hint 8.30 (i)

* Theorem 8.32 — Let $T: V \to V$ be a normal operator on a Hermitian space V,

If (v, λ) is an eigen vector of T $\Rightarrow$ $(v, \bar{\lambda})$ is an eigen vector of T^*

Hint: Consider the operator $\tilde{T} : T - \lambda I$

* Spectral Theorem 8.33 — Let $T: V \to V$ be a normal operator on a Hermitian space V. Then there exists a orthonormal basis $[\mathfrak{B}]$ of V s.t it consists of eigen vectors of T.

Proof: characteristic polynomial $\to$ $\exists (u_1, \lambda)$ an eigen vector of T $\to$ $V = \underbrace{U_1}_{\text{span}(u_1)} \oplus U_1^\perp$

$U_1^\perp$ is T invariant $\leftarrow$ 8.32 + 8.31

$T: U_1^\perp \to U_1^\perp$ is an normal operator $\leftarrow$ Induction of dimension of V.

* Corollary 8.34 (i) If A is a normal matrix, then $\exists$ a unitary matrix P s.t P^*AP is diagonal.

(ii) Every conjugacy class of a normal matrix contains a diagonal matrix.

* Corollary 8.35 Let $T: V \to V$ be a hermitian operator on a Hermitian space V, then the eigen values of T are real.

* Theorem 8.36 Let $T: V \to V$ be a symmetric operator on a euclidean space V, then there exists a orthonormal basis $[\mathcal{B}]$ of V consisting of eigen vectors of T.

Hint: 8.35 & similar steps to 8.33

* Corollary 8.37 If A is a symmetric matrix then $\exists$ a orthonormal matrix P s.t $P = P^TAP$ is diagonal

* Defn 8.38 A bilinear form on a vectorspace over field F is called skew symmetric if

$$\langle v, v\rangle = 0 \quad \forall v \in V \text{ or } \quad \text{— ①}$$

$$\langle u, v\rangle = -\langle v, u\rangle \quad \forall u, v \in V \quad \text{— ②}$$

Conditions ① and ② are same if the characteristic of the field is not 2 if it is 2 then ① is correct.

* Defn 8.39 If $\langle , \rangle$ is a non degenerate skew symmetric bilinear form on a finite dimensional vectorspace V over field F, then $\exists$ a basis $[\mathcal{B}]$ of V s.t the matrix of $\langle , \rangle$ has the structure

$$S_0 = \begin{bmatrix} \xi & & \\ & \xi & \\ & & \ddots \\ & & & \xi \end{bmatrix} \quad \text{where} \quad \xi = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$$

Hint: non degenerate : $\exists \overset{\text{non zero}}{v_1, v_2}$ s.t $\langle v_1, v_2\rangle = c \xrightarrow{\text{scale } v_2} \langle v_1, v_2\rangle = 1$

$\Leftarrow$ v_1 and v_2 are linearly independent $\Leftarrow$ $\langle v_1, v_2\rangle \neq 0$, $\langle v_1, v_1\rangle = 0$

$\Leftarrow$ $V = U \oplus U^\perp$ (U non degenerate, $U^\perp$ non degenerate) $\Leftarrow$ $U = \text{span}(v_1, v_2)$ (nondegenerate)

$\Leftarrow$ Induction on dimension